

## Subject Description Form

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|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subject Code</b>                                   | SFT5106                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Subject Title</b>                                  | Sustainable Production System in Fashion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Credit Value</b>                                   | 3 credits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Level</b>                                          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Objectives</b>                                     | <p>This subject has the following objectives:</p> <ol style="list-style-type: none"> <li>1. To explore the concepts and principles of sustainable production systems in fashion (SPSF).</li> <li>2. To analytically assess the performance of the alternative configurations of EFPSF.</li> <li>3. To analyse the importance of information in SPSF.</li> <li>4. To apply different tools for studying and solving SPSF problems.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Intended Learning Outcomes</b>                     | <p>Students who have successfully completed the subject shall be able to:</p> <ol style="list-style-type: none"> <li>a. Apply analytical tools and models to explore SPSF problems.</li> <li>b. Identify and analyze the challenges in SPSF.</li> <li>c. Propose appropriate analytical solutions for solving the problems in SPSF.</li> <li>d. Present and discuss SPSF issues in English effectively and clearly.</li> <li>e. Co-operate and contribute as a team member/leader.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <p><b>(I) Concepts of sustainable production systems in fashion (SPSF)</b><br/>The need for an integrated production system in the fashion industry. Structural design of SPSF. Selecting and evaluating the alternative configurations of SPSF.</p> <p><b>(II) Environmentally friendly and responsible production</b><br/>Reviews of classical production management models. Managerial principles in SPSF. Ethical production systems.</p> <p><b>(III) Use of Information in SPSF</b><br/>The importance of information in SPSF. Alliances for modern production systems. The applications of information technologies, including block chain and RFID.</p> <p><b>(IV) Circular production systems</b><br/>Differences among reverse logistics production systems, remanufacturing systems and circular production systems. How to establish circular production systems. Circular production in the sharing economy.</p> <p><b>(V) System of systems concept</b><br/>A discussion on the system of systems theory and concept for SPSF.</p> <p><b>(VI) Contemporary Issues in SPSF</b><br/>A discussion on the latest developments and hot topics in SPSF. Case studies in SPSF</p> |
| <b>Teaching/<br/>Learning</b>                         | This subject comprises lectures and tutorials. Students are required to examine contemporary issues in the SPSF, and have group discussions. Simulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------|--------------------------|---|---|---|--|
| Methodology                                                     | games and computerized experiments are used to further illustrate some important concepts. Guided readings will be suggested for private study to complement the subject                                                                                                                                                                                                                                                                                                                                                          |             |                                                                                |                          |   |   |   |  |
| Assessment Methods in Alignment with Intended Learning Outcomes | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |                          |   |   |   |  |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             | a                                                                              | b                        | c | d | e |  |
|                                                                 | 1. Short Assignments/Tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 25%         | ✓                                                                              | ✓                        | ✓ |   |   |  |
|                                                                 | 2. Group projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 25%         |                                                                                | ✓                        | ✓ | ✓ | ✓ |  |
|                                                                 | 3. Final exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 50%         | ✓                                                                              | ✓                        | ✓ |   |   |  |
|                                                                 | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 100%        |                                                                                |                          |   |   |   |  |
|                                                                 | <p>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:<br/>Short assignments (including home-works, class-works, and tutorial exercises), both individual and group-based, and tests are used to assess the first three learning outcomes related to the professional knowledge on SPSF.</p> <p>A comprehensive group project, which includes case analysis and presentation, is employed to assess four learning outcomes, including the two generic outcomes d and e.</p> |             |                                                                                |                          |   |   |   |  |
| Student Study Effort Expected                                   | Class Contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                |                          |   |   |   |  |
|                                                                 | ▪ Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                | 26 hrs. (i.e.; 13*2 hrs) |   |   |   |  |
|                                                                 | ▪ Tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                | 13 hrs. (i.e.; 13*1 hr)  |   |   |   |  |
|                                                                 | Other Study Effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                |                          |   |   |   |  |
|                                                                 | ▪ Assignments + Reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                                | 70 hrs.                  |   |   |   |  |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                                                |                          |   |   |   |  |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                                                |                          |   |   |   |  |
|                                                                 | Total Student Study Effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                | 109 hrs.                 |   |   |   |  |
| Reading List and References                                     | <b>Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                                                                |                          |   |   |   |  |
|                                                                 | Atasu, A. (Ed.) Environmentally Responsible Supply Chains, Springer, 2016.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                |                          |   |   |   |  |
|                                                                 | Choi, T.M., T.C.E. Cheng (Eds.) Sustainable Fashion Supply Chain Management: From Sourcing to Retailing, Springer, 2015.                                                                                                                                                                                                                                                                                                                                                                                                          |             |                                                                                |                          |   |   |   |  |
|                                                                 | Lehmacher, W. The Global Supply Chain: How Technology and Circular Thinking Transform Our Future, 2017.                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                                |                          |   |   |   |  |
|                                                                 | <b>Journal papers:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                |                          |   |   |   |  |
|                                                                 | Cai, Y.J., T.M. Choi. Extended producer responsibility: a systematic review from operations management perspectives. IEEE Transactions on Engineering Management, published online, 2019.                                                                                                                                                                                                                                                                                                                                         |             |                                                                                |                          |   |   |   |  |
|                                                                 | Choi, T.M., Y.J. Cai, B. Shen. Sustainable fashion supply chain management: a system of systems analysis. IEEE Transactions on Engineering Management, 66(4), 730-745, 2019.                                                                                                                                                                                                                                                                                                                                                      |             |                                                                                |                          |   |   |   |  |

Choi, T.M., Y.J. Cai. Impacts of lead time reduction on fabric sourcing in apparel production with yield and environmental considerations. *Annals of Operations Research*, published online, 2018.

Choi, T.M., Y. He. Peer-to-peer collaborative consumption for fashion products in the sharing economy: Platform operations. *Transportation Research – Part E*, 126, 49-65, 2019.

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Choi, T.M., W.K. Yeung, T.C.E. Cheng, X. Yue. Optimal scheduling and the value of RFID technology in garment manufacturing systems. *IEEE Transactions on Engineering Management*, 65(1), 72-84, 2018.

Guo, S., T.M. Choi, B. Shen. Green product development under competition: a study of the fashion apparel industry. *European Journal of Operational Research*, 280(2), 523-538, 2020.

Hosseini-Motlagh, S.M., M. Nouri-Harzvili, T.M. Choi\*, S. Ebrahimi. Reverse supply chain systems optimization with dual channel and demand disruptions: sustainability, CSR investment and pricing coordination. *Information Sciences*, 503, 606-634, 2019.

Li, G., L. Li, T.M. Choi, S.P. Sethi. Green supply chain management in Chinese firms: innovative measures and the moderator role of quick response technology. *Journal of Operations Management*, accepted and in press, 2019.

Sheu, J.B., T.M. Choi. Extended consumer responsibility: syncretic value-oriented pricing strategies for trade-in-for-upgrade programmes. *Transportation Research – Part E*, 122, 350-367, 2019.